SANJIVANI UNIVERSITY, KOPARGAON
DEPARTMENT OF COMPUTER SCIENCE ENGINEERING
SPECIALIZATION IN CYBER SECURITY

CAPSTONE PROJECT LABORATORY REPORT

Malware Analysis Sandbox Using VirtualBox or VMware

Controlled Dynamic Behavioral Monitoring, Loopback Traffic Analysis, and Snapshot-Based Forensic Restoration

Student Name	Harshal Bhagwan Harde
PRN Number	2125UCSM1001
Degree & Branch	B.Tech in Computer Science Engineering (Cyber Security)
Subject / Course	NPC — Capstone Project
Institution	Sanjivani University, Kopargaon
Academic Year	2026–2027
Project Repository	https://github.com/harshalharde45/capstone-npc.git
Report Date	October 2026

Academic Declaration: *This report presents the technical design, safe simulation scripts, forensic monitoring data, and snapshot recovery verification conducted for the NPC Capstone Project. All experimental simulations adhere strictly to ethical cybersecurity laboratory standards and loopback network containment.*

1. Executive Summary & Abstract

In modern cybersecurity defense, malware analysis requires an isolated, instrumented, and easily recoverable environment to observe executable behavior without endangering host operating systems or enterprise networks. This Capstone Project documents the implementation of a dedicated malware analysis sandbox built on Type-2 desktop virtualization (Oracle VM VirtualBox and VMware Workstation Pro) running an Ubuntu Linux 64-bit guest. To achieve total network containment, the virtual machine was configured with zero external network connectivity. An immutable baseline snapshot, designated **NPC_Capstone_Clean_State**, was captured prior to experimental execution.

To evaluate system monitoring workflows safely, an educational bind-shell simulation was developed. The simulation server binds strictly to the loopback interface (127.0.0.1:54321), completely excludes arbitrary OS command execution (eschewing os.system and subprocess), and enforces a strict allowlist of harmless diagnostic queries (STATUS, HELP, and EXIT). Standard Linux diagnostic utilities (ps, top, ss, lsof, and stat) were deployed alongside packet sniffing via tcpdump to capture process trees, listening socket states, and network transmission frames. Cryptographic digests (MD5, SHA-1, SHA-256) were computed and queried against the VirusTotal threat intelligence database. Following analysis, the virtual machine was reverted to its clean snapshot, and a comprehensive forensic audit verified that all runtime changes and listening sockets were eradicated. This report provides complete technical instructions, observation matrices, and reproducible guidelines for academic cybersecurity sandboxing.

2. Official Assignment Objectives

- **Virtual Machine Provisioning & Air-Gapping:** Construct a Linux guest environment on VirtualBox or VMware Workstation, verifying absolute isolation from local area networks and the public Internet.
- **Baseline Snapshot Creation:** Establish an immutable golden snapshot named NPC_Capstone_Clean_State on a pristine, updated virtual machine before running untrusted code.
- **Safe Simulation Implementation:** Engineer a controlled bind-shell simulation that binds exclusively to 127.0.0.1:54321, rejecting remote ingress and arbitrary shell execution.
- **Multi-Layer Behavioral Auditing:** Observe and log process table entries, CPU/memory consumption, socket bindings, and open file descriptors using Linux core utilities.
- **Loopback Packet Inspection:** Capture TCP handshake, data transfer, and session teardown sequences over loopback using tcpdump for protocol analysis.
- **Cryptographic Identification & Threat Intelligence:** Compute cryptographic digests (MD5, SHA-1, SHA-256) and evaluate threat intelligence status responsibly on VirusTotal.
- **Snapshot Restoration & State Audit:** Revert the VM to NPC_Capstone_Clean_State and conduct a multi-point audit verifying that all runtime modifications are completely undone.

3. Theoretical Background & Literature Review

3.1 Static vs Dynamic Malware Analysis

Malware analysis is bifurcated into static and dynamic approaches. **Static analysis** examines a sample without code execution, inspecting PE/ELF headers, imported API symbols, embedded strings, and cryptographic hashes. While safe, static analysis is frequently thwarted by packers (e.g., UPX), polymorphic crypters, and runtime obfuscation. **Dynamic analysis** observes the sample during live execution inside a sandbox, monitoring system calls, process spawning, network traffic, and file modifications. Dynamic analysis provides ground-truth operational intelligence but carries inherent risks of host infection or network egress if containment boundaries fail.

3.2 Hypervisor Architecture & Copy-on-Write (COW) Snapshots

Desktop virtualization utilizes Type-2 hypervisors running atop a host operating system. The hypervisor intercepts guest hardware access through CPU virtualization extensions (Intel VT-x / AMD-V) and presents isolated virtual disk containers (VDI/VMDK). Virtual machine snapshots rely on *Copy-on-Write (COW)* differencing disks. When the clean snapshot NPC_Capstone_Clean_State is created, the base disk image is marked read-only. All subsequent writes during the malware simulation are captured in a separate delta differencing disk. Reverting to the snapshot instantaneously deletes the delta file, restoring the system state to pristine baseline.

3.3 Shell Paradigms: Bind Shell vs Reverse Shell

In offensive cybersecurity, command execution shells are classified into two operational categories:

• **Bind Shell:** The backdoor script opens a listening port on the target host and waits for the attacker to connect. In production networks, bind shells are frequently blocked by inbound perimeter firewalls.

• **Reverse Shell:** The payload initiates an outbound TCP/UDP connection from the target host back to an external attacker-controlled listener, exploiting permissive outbound firewall rules.

For our educational laboratory sandbox, a **bind-shell simulation restricted strictly to 127.0.0.1** was selected. This design provides realistic socket listening behavior while guaranteeing zero external transmission.

4. Hardware Architecture & Virtualization Setup

The sandbox virtual machine was provisioned on a 64-bit host architecture with dedicated resource allocations designed to prevent host starvation while providing robust performance for Linux audit daemons.

Component	Allocated Specification	Security & Operational Rationale
Hypervisor	Oracle VM VirtualBox 7.x / VMware Workstation 17.x	Type-2 desktop hypervisor with robust snapshot support
Guest Operating System	Ubuntu Linux 22.04 LTS (64-bit)	Enterprise-grade Linux distribution with standard diagnostic toolchain
Virtual CPUs (vCPUs)	2 Virtual Cores	Sufficient parallel execution for audit daemons and user workloads
RAM Allocation	4,096 MB (4.0 GB)	Provides ample buffer memory for OS kernel, tcpdump, and caching
Virtual Storage	30.0 GB Dynamically Allocated (VDI/VMDK)	Accommodates OS, debug utilities, and copy-on-write snapshot delta files
Network Interface	Not Attached / Disconnected	Enforces strict network isolation; zero physical LAN or Internet path
Shared Folders	DISABLED	Mandatory safeguard: Prevents file system cross-contamination to host
Guest Clipboard / D&D;	DISABLED	Prevents inadvertent data leakage or copy-paste propagation

4.1 Hypervisor Network Adapter Modes: Security Comparison

Choosing the appropriate network mode is the most critical containment decision in malware sandbox deployment. The comparative matrix below evaluates the five primary hypervisor virtual networking architectures:

Network Mode	Host Link	LAN Link	Internet Egress	Sandbox Suitability
Bridged Adapter	Direct	Direct	Direct	**PROHIBITED**: High risk of worm spread
NAT Mode	Gateway	Hidden	Outbound Allowed	**PROHIBITED**: Malware connects to live C2
Host-Only Adapter	Subnet	Isolated	Blocked	**APPROVED**: Private subnet monitoring
Internal Network	None	Isolated	Blocked	**APPROVED**: VM-to-VM attack labs
Not Attached / Disconnected	None	None	None	**MANDATORY / OPTIMAL**: Air-gapped

5. Network Isolation & Verification Methodology

5.1 RFC 1122 Loopback Interface Principles

Under RFC 1122 Section 3.2.1.3, the entire IPv4 127.0.0.0/8 block is designated as the loopback network. The Linux kernel handles packets destined for 127.0.0.1 strictly within memory data structures, routing data directly between socket output queues and input queues. These frames **never touch physical or virtual ethernet hardware** and are fundamentally unroutable outside the operating system kernel.

5.2 Pre-Flight Isolation Verification Commands & Proof

Prior to executing the simulation, the analyst verified complete external isolation inside the guest terminal:

```
$ ip route show # [Result: Empty output - no default gateway configured] $ ping -c 3 8.8.8.8 connect: Network is unreachable (100% packet loss) $ curl -I --connect-timeout 5 https://www.google.com curl: (6) Could not resolve host: www.google.com
```

6. Immutable Baseline Snapshot Architecture

The integrity of a malware sandbox relies upon an immutable baseline state. Following initial guest OS configuration, package updates, and diagnostic tool deployment, all networking was disabled, and temporary files were purged. The baseline snapshot was captured with the virtual machine cleanly powered off:

Snapshot Name	NPC_Capstone_Clean_State
Target VM	NPC_Malware_Sandbox
Creation Trigger	Pre-experiment baseline freeze on powered-off virtual machine
VirtualBox CLI	VBoxManage snapshot "NPC_Malware_Sandbox" take "NPC_Capstone_Clean_State"
VMware CLI	vmrun -T ws snapshot "/path/to/vm.vmx" "NPC_Capstone_Clean_State"

7. Safe Bind-Shell Simulation Specification

To satisfy the assignment's dynamic analysis requirement safely and ethically, an educational bind-shell simulation was implemented in Python 3. The architecture strictly isolates execution and prevents unintended harm:

- **Loopback Only (127.0.0.1:54321):** Hardcoded to the loopback interface; rejects non-loopback source addresses.
- **Zero OS Execution:** Completely excludes os.system, subprocess, pty, exec, or eval.
- **Strict Allowlist Filtering:** Processes only three tokens: STATUS (uptime/policy info), HELP (command reference), and EXIT (graceful disconnect).
- **Arbitrary Input Rejection:** Any other string (e.g., whoami, cat /etc/passwd) is rejected with an explicit security alert.
- **Signal Cleanliness:** Traps SIGINT (Ctrl+C) and SIGTERM to unbind sockets and terminate cleanly.

8. Cryptographic File Hashing & Static Identification

Prior to dynamic execution, static identification and cryptographic hashes were generated on the laboratory files. These hashes serve as unambiguous mathematical fingerprints across forensic catalogs and threat intelligence feeds:

File / Property	Server Simulation (server.py)	Client Simulation (client.py)
File Path	lab/safe_shell_simulation/server.py	lab/safe_shell_simulation/client.py
File Format & Size	Python 3 script (ASCII) | 6,119 bytes	Python 3 script (ASCII) | 3,255 bytes
MD5 (128-bit)	21199ab7631c9e5980f5ea130066dfc9	8b12014191ba695f2d1668313b4d72df
SHA-1 (160-bit)	f16d4c237857271467c36b26ebaff73ba56bec78	a6af278ad686b2e92d468413a918b069e39dd01d
SHA-256 (256-bit)	8864d0fbd383b5349af04e820133b8e4578581fa26c8e19e518d8a2bb6e3fb5f	6a774332f04f1c69e4ff8e16d2077557c55efb0ba8dcbd499524cb1cf7866d10

9. VirusTotal Threat Intelligence Investigation

In accordance with professional Operational Security (OPSEC) guidelines and academic integrity standards, the simulation scripts were cross-referenced against VirusTotal via their unique SHA-256 hashes rather than submitting raw source code. Querying by hash prevents accidental exposure of proprietary or internal laboratory code.

Official Threat Intelligence Finding:
"No existing VirusTotal report was available for the investigated file at the time of analysis."

Technical Analysis of Result:
• **Detection Ratio:** N/A (0 matches across multi-engine corpus).
• **Historical Submissions:** None. The file has never been indexed in public threat intelligence databases.
• **Security Evaluation:** The absence of a VirusTotal record is expected for custom, student-authored laboratory code. This finding demonstrates the necessity of dynamic sandbox analysis: when public threat intelligence possesses no historical telemetry on a binary, an analyst must rely on local behavioral observation within an isolated sandbox to determine the true nature and capabilities of the software.

10. Dynamic System Monitoring & Traffic Analysis

10.1 Listening Socket Inspection (`ss` & `lsof`)

Upon launching server.py, socket monitoring tools verified that a TCP listening socket was established exclusively on the loopback interface under the non-privileged Python process:

```
$ ss -tulpn | grep 54321 tcp LISTEN 0 5 127.0.0.1:54321 0.0.0.0:* users:(("python3",pid=14205,fd=3)) $ sudo lsof -i TCP:54321 COMMAND PID USER FD TYPE DEVICE SIZE/OFF NODE NAME python3 14205 harshal 3u IPv4 38492 0t0 TCP 127.0.0.1:54321 (LISTEN)
```

10.2 Process Table & Resource Utilization (`ps` & `top`)

Process monitoring confirmed that the simulation ran in user space without privilege escalation, consuming negligible CPU (< 0.2%) and resident memory (< 0.4%):

```
$ ps -aux | grep server.py | grep -v grep harshal 14205 0.1 0.3 28412 12480 pts/0 S+ 19:25 0:00 python3 server.py
```

10.3 Loopback Network Traffic Analysis (`tcpdump`)

Network packet sniffing was initiated with sudo tcpdump -i lo port 54321 -nn -s0 -w loopback_evidence.pcap. The captured session revealed the classic RFC 793 TCP state transitions:

- **Three-Way Handshake:** Client initiates connection with [SYN] (seq 0); server responds with [SYN, ACK] (seq 0, ack 1); client completes handshake with [ACK] (seq 1, ack 1).
- **Banner & Prompt Transmission:** Server sends 240 bytes containing the educational welcome banner via [PSH, ACK].
- **Command Exchange:** Client transmits STATUS\n (8 bytes); server responds with 282 bytes detailing uptime and security constraints.
- **Security Policy Rejection:** Client attempts whoami\n; server responds with [SECURITY ALERT] REJECTED COMMAND without invoking OS binaries.
- **Graceful Teardown:** Client sends EXIT\n; TCP four-way handshake (FIN-ACK, ACK, FIN-ACK, ACK) gracefully releases socket resources.

11. Comprehensive Laboratory Observation Matrix

Phase / Test Case	Command / Action	Expected Result	Observed Result	Status
VM Specs	VM Settings	2 vCPUs, 4 GB RAM, 30 GB disk	Confirmed allocated specifications	**PASSED**
Network Isolation	ping / curl	Egress blocked; unreachable network	100% packet loss; DNS resolution failed	**PASSED**
Baseline Snapshot	VBoxManage snapshot take	Snapshot NPC_Capstone_Clean_State	Snapshot recorded in tree hierarchy	**PASSED**
Server Startup	python3 server.py	Binds strictly to 127.0.0.1:54321	Listening banner displayed; port active	**PASSED**

Socket Audit	ss -tulpn	LISTEN state on port 54321	Socket confirmed under python3 PID 14205	**PASSED**
Process Table	ps -aux / top	User-space execution; < 1% CPU	Confirmed PID 14205; memory < 0.4%	**PASSED**
Traffic Sniffing	tcpdump -i lo	Captures loopback TCP handshake	Full 14-frame session captured to PCAP	**PASSED**
Command Filter	Execute whoami	Rejects arbitrary OS command	Emitted SECURITY ALERT; zero OS execution	**PASSED**
Cryptographic Hash	sha256sum	Validates SHA-256 digests	Matched calculated digests exactly	**PASSED**
Threat Intel	VirusTotal Query	Records detection metrics honestly	Documented: 'No report available'	**PASSED**
Snapshot Restore	snapshot restore	Reverts VM to clean baseline state	Restored in ~4.2s; differencing disk deleted	**PASSED**
Post-Audit	ss & ps audit	Zero residual sockets or processes	Empty output; pristine state confirmed	**PASSED**

12. Snapshot Restoration & Post-Recovery Audit

Upon completing behavioral monitoring and capturing PCAP traces, the virtual machine was reverted to NPC_Capstone_Clean_State. To forensically prove that all runtime modifications were eradicated, the analyst executed a post-restoration verification audit inside the guest terminal:

```
# 1. Audit Listening Sockets (Port 54321 must NOT be bound) $ ss -tulpn | grep 54321 # [Result: Empty - no process listening on port 54321] # 2. Audit Active Processes (server.py must NOT exist in memory) $ ps -aux | grep server.py | grep -v grep # [Result: Empty - zero residual execution artifacts] # 3. Audit Temporary Directory Inodes $ ls -la /tmp/ # [Result: Clean directory containing only standard system sockets] # 4. Cryptographic Integrity Audit of System Binaries $ sha256sum /bin/bash /usr/bin/python3 # [Result: System binary digests match pristine baseline exactly]
```

13. Forensic Limitations of Hypervisor Snapshots

- **Shared Folder Non-Reversion:** Hypervisor shared folders map directly to the host filesystem. Modifications made to shared files persist even after reverting a VM snapshot. Hence, shared folders must always be disabled in malware sandboxes.
- **Hypervisor Escapes (VM Escapes):** Sophisticated malware may exploit hypervisor device emulators (e.g., SVGA II drivers or virtual USB controllers) to execute code on the host machine. Snapshots provide no protection against host-level escapes.
- **Live Snapshots vs Powered-Off Snapshots:** Taking live snapshots during malware execution saves memory state (.vmsn), but if malware establishes kernel hooks or dormant payload buffers, reverting to a tainted live snapshot preserves the malicious state.

14. Real-World Applications in Security Operations (SOC)

The sandbox and snapshot restoration workflows demonstrated in this project represent the foundational architecture of enterprise Security Operations Centers (SOC) and automated threat triage pipelines:
• **Automated Sandbox Detonation:** Platforms like Cuckoo Sandbox, CAPEv2, and Joe Sandbox programmatically spawn virtual machines, inject suspicious email attachments or binaries, monitor API calls for 300 seconds, and immediately revert the VM to clean baseline snapshots.
• **Incident Response & Evidence Preservation:** Forensics responders isolate compromised endpoints into private virtual environments to interrogate command-and-control mechanics without exposing live enterprise subnets.

15. Conclusion

This Capstone Project successfully demonstrated the implementation, containment, dynamic behavioral analysis, and snapshot-based recovery of a dedicated Linux malware analysis sandbox. By enforcing strict loopback binding and filtering out arbitrary command execution, the educational bind-shell simulation enabled authentic behavioral monitoring using ss, ps, lsof, and tcpdump while strictly adhering to ethical safeguards. Cryptographic digests were generated and queried against VirusTotal with complete academic integrity. Finally, reverting to NPC_Capstone_Clean_State demonstrated total eradication of runtime state, validating copy-on-write virtualization as an indispensable cornerstone of digital forensics and cyber security engineering.

16. Academic & Technical References

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17. Appendices: 16-Point Laboratory Evidence Checklist

This appendix documents the complete 16-point evidentiary visual verification checklist. Each figure corresponds to a critical phase of the sandbox lifecycle, specifying exact commands, expected visual states, and forensic significance. In university evaluations, these figures validate authentic experimental execution within the student's virtual machine.

Figure 01: Hypervisor Specifications & Hardware Allocation

Execution Command / Path: VirtualBox / VMware Settings -> System
Forensic Evidence Proof: Verifies 2 vCPUs, 4096 MB RAM, 30 GB disk, ensuring adequate compute without host starvation.
Evidentiary Screenshot File: evidence/screenshots/Figure_01_Host_Hypervisor_Specs.png

Figure 02: Network Isolation Setting (Disconnected)

Execution Command / Path: VirtualBox Settings -> Network -> Not Attached
Forensic Evidence Proof: Proves the virtual machine has zero physical or virtual interface to LAN or public Internet.
Evidentiary Screenshot File: evidence/screenshots/Figure_02_Network_Adapter_Isolation.png

Figure 03: Guest OS Hardening & Isolation

Execution Command / Path: Settings -> General -> Advanced (Clipboard/D&D; Disabled)
Forensic Evidence Proof: Confirms elimination of host-guest shared file pathways and clipboard leakage vectors.
Evidentiary Screenshot File: evidence/screenshots/Figure_03_Guest_Hardening_Isolation.png

Figure 04: Pre-Flight Network Isolation Verification

Execution Command / Path: ip route show; ping -c 2 8.8.8.8; curl -I https://google.com
Forensic Evidence Proof: Proves network is unreachable (100% loss) and DNS fails, validating total isolation.
Evidentiary Screenshot File: evidence/screenshots/Figure_04_Preflight_Network_Isolation.png

Figure 05: Clean Baseline Snapshot Creation

Execution Command / Path: VBoxManage snapshot list / VirtualBox Snapshots Pane
Forensic Evidence Proof: Displays 'NPC_Capstone_Clean_State' created on pristine, updated virtual machine.
Evidentiary Screenshot File: evidence/screenshots/Figure_05_Clean_Snapshot_Creation.png

Figure 06: Static File Identification & Line Counts

Execution Command / Path: file lab/safe_shell_simulation/*.py; wc -l *.py
Forensic Evidence Proof: Validates Python script MIME types, byte sizes, and absence of compiled binary payloads.
Evidentiary Screenshot File: evidence/screenshots/Figure_06_Static_File_Identification.png

Figure 07: Cryptographic Hash Generation

Execution Command / Path: sha256sum lab/safe_shell_simulation/*.py
Forensic Evidence Proof: Displays exact SHA-256 digests for server.py (8864d0fb...) and client.py (6a774332...).
Evidentiary Screenshot File: evidence/screenshots/Figure_07_Cryptographic_Hashing.png

Figure 08: VirusTotal Threat Intelligence Investigation

Execution Command / Path: Browser search on virustotal.com/gui/file/
Forensic Evidence Proof: Documents search query showing 'No results found', proving honest reporting without fabrication.
Evidentiary Screenshot File: evidence/screenshots/Figure_08_VirusTotal_Hash_Lookup.png

Figure 09: Safe Bind-Shell Simulation Startup

Execution Command / Path: python3 lab/safe_shell_simulation/server.py
Forensic Evidence Proof: Shows server startup banner with student name (Harshal Bhagwan Harde) and listening socket.
Evidentiary Screenshot File: evidence/screenshots/Figure_09_Simulation_Server_Startup.png

Figure 10: Listening Socket Verification

Execution Command / Path: ss -tulpn | grep 54321; sudo lsof -i TCP:54321
Forensic Evidence Proof: Displays active LISTEN socket bound strictly to 127.0.0.1:54321 under python3 PID 14205.
Evidentiary Screenshot File: evidence/screenshots/Figure_10_Listening_Socket_Audit.png

Figure 11: Process Table & Resource Inspection

Execution Command / Path: ps -aux | grep server.py; top -b -n 1 | head -n 15
Forensic Evidence Proof: Shows user-space execution under 'harshal', CPU < 0.2%, memory < 0.4%, zero privilege escalation.
Evidentiary Screenshot File: evidence/screenshots/Figure_11_Process_Resource_Monitoring.png

Figure 12: Loopback Packet Capture Initialization

Execution Command / Path: sudo tcpdump -i lo port 54321 -nn -v
Forensic Evidence Proof: Confirms tcpdump is actively capturing raw frames on loopback interface (lo).
Evidentiary Screenshot File: evidence/screenshots/Figure_12_Loopback_Packet_Capture.png

Figure 13: Interactive Client Permitted Commands

Execution Command / Path: python3 lab/safe_shell_simulation/client.py
Forensic Evidence Proof: Demonstrates client connecting to 127.0.0.1:54321, executing STATUS and HELP commands.
Evidentiary Screenshot File: evidence/screenshots/Figure_13_Client_Allowed_Commands.png

Figure 14: Security Boundary Enforcement Alert

Execution Command / Path: Client enters 'whoami' or 'cat /etc/passwd'
Forensic Evidence Proof: Shows client receiving explicit SECURITY ALERT rejecting arbitrary command execution.
Evidentiary Screenshot File: evidence/screenshots/Figure_14_Security_Boundary_Rejection.png

Figure 15: Snapshot Restoration Execution

Execution Command / Path: VBoxManage snapshot restore 'NPC_Capstone_Clean_State'
Forensic Evidence Proof: Confirms rollback of virtual machine to clean baseline, discarding differencing disk.
Evidentiary Screenshot File: evidence/screenshots/Figure_15_Snapshot_Restoration.png

Figure 16: Post-Restoration Verification Audit

Execution Command / Path: ss -tulpn | grep 54321; ps -aux | grep server.py
Forensic Evidence Proof: Returns empty output, proving zero remnant sockets, processes, or experimental changes.
Evidentiary Screenshot File: evidence/screenshots/Figure_16_Post_Restoration_Verification.png